



BRAKE POWER CERTIFICATE (BPC)

1. INTRODUCTION

- Before a train is offered for service, **every coach/wagon must be examined** for mechanical and brake fitness.
 - This examination is done by **C&W staff**.
 - After examination, a formal **Brake Power Certificate (BPC)** is issued to authorize the train to run.
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2. WHAT IS BPC?

- **A Safety Document** that certifies the:
 - **Brake continuity**
 - **Brake power percentage**
 - **Mechanical fitness** of the train
 - Ensures that the train meets the **minimum brake power requirements** and is **safe to run** on the section.
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3. WHO ISSUES & SIGNS THE BPC?

Issued by

- **SSE/IE (C&W)** at nominated examination points.

Joint Signatures Required

- **TXR (Train Examiner / C&W staff)**
- **Guard (Train Manager)**
- **Loco Pilot (LP)**

These signatures together confirm **complete brake continuity** and **fitness** of the rake.

4. FUNCTIONS / IMPORTANCE OF BPC

- Certifies **minimum Brake Power %** for the train (varies by type).
- Ensures:

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- Proper coupling of CBCs & air hoses.
- Brake pipe continuity.
- Mechanical integrity.
- Train has no hot axle / loose fittings.
- Serves as an **official record** for safe running.

TYPES OF TRAINS & THEIR BPC VALIDITY (FREIGHT TRAIN EXAMINATION)

1. TYPES OF FREIGHT EXAMINATIONS

Only 3 types:

- **CC Rake Examination**
- **Premium Rake Examination**
- **End-to-End Examination**

2. CLOSED CIRCUIT (CC) RAKES

What is a CC Rake?

- Periodical Monitoring Examination (PME) done only at **base depot**.
- Runs in fixed **region of operation** (Zones only, not station-specific).
- Integrity must be maintained.

Validity of BPC (CC)

- **7500 km or 35 days**

Key Rules

- BPC issued **only at base depot**.
- After each loading/unloading → **GLP revalidation mandatory**.
- LP must **log distance** in BPC every trip.
- If km data missing & not retrievable → BPC **valid only for 20 days**.

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- **Color of BPC: Yellow**

Overdue CC Rakes Handling

- If empty → move to base depot with GLP check.
- If loaded → GLP up to nearest TXR point; endorse for movement up to destination → then empty to depot
- POH/ROH wagons need not be detached unless unsafe.

When BPC becomes INVALID (CC)

- Integrity disturbed by >4 wagons.
- Stabled >24 hrs at TXR yard.
- Runs outside assigned Zones.
- Not moving toward PME depot when overdue.
- Exceeds **7500 km / 35 days**

Close Circuit (CC) Rakes

(Air brake stock only)

Validity

- **7500 km / 35 days** (whichever earlier)

Minimum Brake Power

- Originating: **100%**
- En-route: **90%**

3. PREMIUM RAKES

Characteristics

- Formed **only in empty condition** at nominated 'A' category depots.
- Must be examined at specified depots (RDM, BPA, SNF, BZA, COA, GY, BTTR, NLPD, PAU).
- Minimum **95% Brake Power**.

Validity

- **12 days** from BPC issue
- Grace period: **+3 days** if rake is loaded on 12th day → must reach TXR yard by 15th day.

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Important Exam Points

- Multiple load/unload allowed within validity.
- After each loading/unloading → **LP + Guard GLP check**
- Last loading must be before 12th day.
- No bypassing of exam points allowed.
- **Color of BPC:** Green

Invalid BPC Conditions (Premium)

- Integrity disturbed >4 wagons.
- Stabled >24 hrs at TXR yard.
- Loading attempted after 12th day.
- Moving loaded after 12 days without examination

Premium Air Brake Rakes

(BOXN, BCN, BRN, BOBRN, etc.)

Validity

- **12 days** for multiple loading
- **+3 days grace** (only if rake remains loaded)

Minimum Brake Power

- Originating: **≥95%**
- En-route: *Not specified*

Key Conditions

- Examination **only in empty condition**
- BPC issued **only from nominated 'A' category depot**

4. END-TO-END RAKES

Applicability

- All rakes **not** in CC or Premium category.

Validity Rules

- Empty rake examined → BPC valid:
 "Up to loading point & further upto unloading point"

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- After loading:
 - BPC must clearly mention **destination**.
 - LP must NOT start train without destination entry.

Critical Time Rule

- Empty rake must reach loading point **within 4 days** of BPC issue.
Otherwise → fresh BPC required.

If unloading point is not a TXR point

- GLP check permitted up to **next TXR point**
- Maximum **400 km** allowed on GLP

Color of BPC

- **Green**

Minimum Brake Power

- **90%**

Invalid BPC (End-to-End)

- Integrity disturbed by >4 wagons.
- Stabled >24 hours at TXR yard.
- Destination not mentioned/incorrect.

Ordinary End-to-End Rakes

(Vacuum brake + Air brake, other than Premium)

Validity

- **One-time loading only**
- Valid from: *Examination* → *Loading* → *Unloading*

Minimum Brake Power

- **Originating station:**
 - 85% (Vacuum brake stock)
 - 90% (Air brake stock)
 - **En-route:** 75%
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5. GLP CHECK (Guard–LP Joint Check)

When GLP is required?

- After every loading/unloading for CC & Premium
- After tripling where no C&W staff
- For end-to-end rakes when unloading station is not TXR point
- Back-loaded trains from non-C&W stations
- Stabled >24 hrs
- When BPC is invalid

Checklist (15 items)

Includes:

- Integrity intact (≤ 4 wagons disturbed)
- All CBCs/hoses coupled
- Angle cocks open (last one closed)
- No hanging/loose parts
- Springs, pads OK
- Doors secured
- No hot axle symptoms
- Continuity of brake pipe pressure confirmed (whistle/VHF)

 **Joint Memo must be prepared in triplicate** and endorsed by SM/YM.
Train cannot start without this memo.

6. BPC REVALIDATION BY SSE/JE (C&W)

When required?

- Post-loading examination
- Post-tripling examination
- Special cases like military specials
- Intensive examination of overdue rakes

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POST-LOADING / POST-TIPPLING EXAM

- **Non-steel loads:** GLP check by Guard & LP
- **Steel loads:** Only SSE/JE (C&W) can certify

UNDER TRIAL ITEMS

- CC BOXN/BOXNHL validity: **10,000 km or 35+5 days**
- Special Premium BCN/BCNA: **20+5 days**
- CC BCN: **9000 km or 30+5 days**
- CC Container: **9000 km or 30 days**
- CC BOBRN: **7500 km or 35+5 days**
- Depot twinning for CC:
 - BOXN: RDM, BPA, BZA, GY
 - BCN: BZA & GY

EXAM-ORIENTED MUST-WRITE POINTS

- “No freight train shall run with invalid BPC.”
- “GLP check is NOT a substitute for BPC; it is a permissive movement tool.”
- “Rake integrity disturbance >4 wagons → BPC invalid.”
- “CC rakes move only within approved Zones.”
- “400 km cap on GLP movements.”
- “Yellow BPC = CC, Green BPC = End-to-End.”

ENHANCED BPC VALIDITY (WAGON-WISE)

A. Close Circuit (CC) Examination Rakes

Wagon Type	Enhanced Validity
BOXN / BOXC / BOXNHL25T / BOYEL / BOYEL / BOYEL variants	10000 km OR 35+5 days (earlier)
BCN / BCNHL	9000 km OR 30+5 days

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Wagon Type	Enhanced Validity
CONTAINER (BLC/BLCA/BLCB)	9000 km OR 30+5 days
BOBRN	7500 km OR 35+5 days
BTPGLN	10000 km OR 35+5 days
BOBSN	8000 km OR 35+5 days
BOYEL	9000 km OR 35+5 days

B. Premium Examination

Category	Enhanced Validity
Steel Rakes (BOST, BRNA, BFNS/BFNSM, BRN22.9)	12+3 days

C. Special Premium Examination

Category	Enhanced Validity
BCN / BCNA Special Premium Rakes	20+5 days

◆ 4. CRITICAL CONDITIONS (EXAM-FOCUSED)

These conditions apply across all trial periods:

1. No loading in expired BPC (KM-wise)

If **KM validity expires** → **NO LOADING**, even if days remain.

2. One-time loading relaxation

Applicable only for:

- **BOXN & its variants**
- Condition:
 - Days validity expired
 - **KM validity < 9000 km**

3. Mandatory Monitoring by Zonal Railways

- Close monitoring of rakes during trial
- Wagon-wise feedback to Railway Board (Annexure A/B)

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BPC: SUMMARY

Category	Validity	BP%	Key Trigger for Invalidity
Ordinary	One-time loading	85/90 → 75	4 days to reach loading point
Premium	12 days + 3	≥95	Not examined at A-depot / >12 days loading
CC	7500 km / 35 days	100 → 90	Not in circuit / >4 wagon disturbance
Container	6000 km/30 days	100 → 90	Out of circuit / km exceeded

MAINTENANCE CATEGORIES (Mainline Coaching Trains)

BRAKE POWER CERTIFICATE (BPC) – COACHING

MAIL / EXPRESS TRAINS — PIT LINE EXAMINED

ICF Coaching Stock

- **Validity:** 3500 km OR 96 hours (4 days)
- Whichever is **earlier**

LHB Coaching Stock

- **Validity:** 4000 km OR 96 hours (4 days)
- Whichever is **earlier**

2 PASSENGER TRAINS (With / Without Toilets)

A. Passenger Trains With Toilets (ICF Stock)

- **Validity: 3500 km OR 96 hours (4 days)**
- Whichever is **earlier**

B. Passenger Trains Without Toilets (ICF Stock)

- **Validity: 3500 km OR 7 days**
 - Whichever is **earlier**
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3 MILITARY / ELECTION SPECIAL TRAINS

ICF Coaches

- **Validity: 3500 km OR 7 days**
- Whichever is **earlier**

LHB Coaches

- **Validity: 4000 km OR 7 days**
 - Whichever is **earlier**
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4 PARCEL EXPRESS (DEDICATED PARCEL TRAINS)

- **Validity: 4500 km OR 10 days**
- Whichever is **earlier**

(Applicable to fully dedicated parcel rakes only)

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MEMU / EMU TRAINS

MEMU Trains with 700 HP Motor Coaches

- **Validity: 7 days**

MEMU Trains with Motor Coaches < 700 HP

- **Validity: 10 days**